AICTE REGULATORY ENGINEERING SPECIFICATION · v2.0

Lattice: Complaint & Risk Intelligence

Complete Technical Architecture, Mathematical Models, RBAC Security & API Reference

Document Property	Specification Value
System Name	Lattice — AICTE Complaint & Institutional Risk Intelligence Platform
Runtime Stack	React 19 + TypeScript + Vite 8 SPA | Express 4 + Node.js Backend (Port 3000)
Authentication & RBAC	Salted SHA-256 Password Auth · Strictly 4 Roles (Chairman, Bureau Head, Case Officer, Admin)
Intelligence Engines	TF-IDF Cosine Similarity + Union-Find Clustering + 5-Factor Risk Formula + Logistic SLA Model
Seeded Dataset	Deterministic Mulberry32 PRNG (Seed 20260419): 40 Institutions, 320 Complaints, 4 Units

SECTION 01

System Overview & Full-Stack Architecture

Lattice is an internal regulatory intelligence platform engineered for the All India Council for Technical Education (AICTE). It unifies complaint intake, automated unit routing, cross-bureau duplicate and context detection, explainable institutional risk scoring, SLA breach forecasting, and strict role-based access control (RBAC) across AICTE bureaus and cells.

1.1 Technology Stack

- **Frontend Application:** React 19.0, TypeScript 7, React Router v7, Tailwind CSS v4 paired with a custom editorial design system (Warm Paper #F6F4EF, Deep Charcoal Ink #15140F, Vermilion Signal #FF5B1F).
- **Typography & Data Readability:** Newsreader (editorial display serif for headings and KPI numerals), Plus Jakarta Sans (UI controls and body copy), and JetBrains Mono (tabular numerals, case reference IDs, similarity scores, and formulas).
- **Motion & Custom SVG Graphics:** GSAP 3.12 (gsap.context lifecycle management, staggered card reveals, animated risk factor bars, and SVG stroke-dashoffset path drawing) with full prefers-reduced-motion compliance.
- **Backend Server & Storage:** Express 4 server (server.ts) mounting REST endpoints (/api/*) and Vite middleware in development. Persistent JSON state is stored at .data/lattice_state.json and automatically initialized with a deterministic 320-complaint seed.

1.2 High-Level Source Tree

```
/server.ts                          -> Express API, RBAC enforcement, TF-IDF & Risk engines,
Seed generator
/src/App.tsx                        -> Root application shell, ToastProvider, BrowserRouter,
session sync
/src/routes.tsx                     -> Role-guarded route definitions (/ , /complaints,
/link-queue, etc.)
/src/api/types.ts                   -> End-to-end TypeScript interfaces for API payloads & RBAC
permissions
/src/api/client.ts                  -> Fetch wrapper with Bearer token injection &
sessionStorage persistence
/src/pages/Login.tsx                -> Password authentication UI, 4-role presets, Section 3
capability matrix
/src/pages/Overview.tsx             -> Chairman Dashboard (national) & Bureau Dashboard (officer
workload)
/src/pages/Complaints.tsx           -> Paginated case registry, multi-facet filters, Case
Officer priority sort
/src/pages/ComplaintDetail.tsx      -> Dossier view, SLA gauge, linked cases, role-gated actions
& redaction
/src/pages/NewComplaint.tsx         -> Complaint intake form with live 300ms debounced
intelligence preview
/src/pages/LinkQueue.tsx            -> Institution-link resolution queue for Bureau Head & Admin
roles
/src/pages/Institutions.tsx         -> Searchable directory of 40 AICTE-approved technical
institutions
/src/pages/InstitutionDetail.tsx    -> Explainable 5-factor risk breakdown, 12-week history &
link topology
/src/pages/Alerts.tsx               -> Rule-based early-warning feed (Rules R1–R4) with
acknowledgement workflow
```

SECTION 02

Roles, Permissions & Password Authentication (Section 3)

The platform implements email and password authentication backed by salted SHA-256 password hashes and strictly enforces four system roles without subroles: chairman, bureau_head, case_officer, and admin.

2.1 Section 3 Role & Capability Matrix

Capability	Chairman	Bureau Head	Case Officer	Admin
Register complaint	no	yes	yes	no
View case list & detail	all bureaus, full	full	full (assigned 1st)	metadata only (3.1)
Assign / reassign officer	no	yes	no	yes
Add action note	no	yes	assigned cases	no
Status transitions	reopen only	per section 5	per section 5	none
Resolve link queue	no	yes	no	yes
Chairman dashboard	yes	no	no	no
Bureau dashboard	no	yes	no	no

2.2 Official Role Credentials

Role	Official Email	Password	Default Landing Route & Key Behavior

Chairman	chairman@aicte.gov.in	Chairman@123	/ (Chairman Dashboard) · Reopen-only transitions
Bureau Head	bureau_head@aicte.gov.in	Bureau@123	/ (Bureau Dashboard) · Full operational control
Case Officer	case_officer@aicte.gov.in	Officer@123	/complaints · Assigned cases sorted first
Admin	admin@aicte.gov.in	Admin@123	/complaints · Metadata-only view, assign & link queue

2.3 Section 3.1 Admin Metadata-Only Redaction

When authenticated as Admin, the backend API (buildComplaintDetail and GET /api/complaints) enforces Section 3.1 privacy rules: the complaint narrative description is returned as null, keyword search excludes narrative bodies, and all officer investigation notes (event.type === "Note") have their message replaced with "[Action note content redacted — Admin metadata-only access]" while preserving routing, SLA, and assignment metadata.

SECTION 03

Core Intelligence Engines & Mathematical Models

3.1 Automated Intake Classification & Statutory SLA Windows

During complaint intake (POST /api/complaints/analyze and POST /api/complaints), text is scored against a weighted regulatory lexicon across 8 categories mapped to 4 AICTE units:

Regulatory Category	Routed Unit	Primary Lexicon Triggers
Fee and Refund	SGC (Grievance)	refund, tuition fee, caution deposit, overcharging, capitation
Document and Certificate Issues	SGC (Grievance)	original degree, marksheet withheld, transfer certificate, transcript
Harassment and Ragging	SGC (Grievance)	ragging, hostel harassment, intimidation, abuse, discrimination
Faculty and Staffing	ACB (Academic)	faculty vacancies, salary unpaid, ghost faculty, professor shortage
Curriculum and Examination	ACB (Academic)	syllabus unapproved, exam delay, grading irregularity, attendance
Infrastructure and Safety	AB (Approval)	electrical hazard, fire safety, laboratory shortage, building seepage
Admissions and Affiliation	AB (Approval)	unapproved intake, excess seats, affiliation renewal, counselling
Placement and Misrepresentation	VC (Vigilance)	fake placement, 100% placement claim, misleading advertisement

```
// Classifier Confidence & Statutory SLA Window Calculation
confidence = clamp(0.55 + 0.08 * bestScore + 0.05 * (bestScore - secondScore), 0.55, 0.96)
SLA_DAYS   = { Critical: 3, High: 7, Medium: 15, Low: 30 }
due_at     = created_at + SLA_DAYS[severity] * 86400 seconds
```

3.2 TF-IDF Cosine Similarity & Union-Find Case Clustering

For every institution, all complaints are tokenized into unigrams and bigrams after stopword removal. The engine builds sublinear TF-IDF vectors and computes pairwise cosine similarity for complaints within the same institution:

```
tfWeight(tf)     = 1 + ln(tf)
idfWeight(df, N) = ln((N + 1) / (df + 1)) + 1
cosine(A, B)     = dot(vecA, vecB) / (||vecA|| * ||vecB||)

// Pairwise Link Classification Thresholds:
1. Duplicate Link : cosine(A, B) >= 0.52
2. Similar Link   : 0.28 <= cosine(A, B) < 0.52
3. Context Link   : unit(A) !== unit(B) AND |created(A) - created(B)| <= 45 days
```

Connected components across Duplicate and Similar edges are merged using a Disjoint-Set (Union-Find) structure to assign a deterministic cluster_id to each complaint.

3.3 5-Factor Composite Institution Risk Score (0–100)

Each institution is evaluated over a rolling 90-day window using an explainable 5-factor linear combination (30V + 25S + 20R + 15A + 10X):

```
V (Volume, weight 0.30)          = min(1.0, count_90d / 12)
S (Severity Mix, weight 0.25)    = sum(sev_weight) / (4 * count_90d)  [Low=1, Med=2, High=3,
Crit=4]
R (Recurrence, weight 0.20)      = min(1.0, ((max_cluster - 1) / 4) * 0.6 + repeat_cat_share *
0.4)
A (SLA Breach Rate, weight 0.15) = min(1.0, (breached_or_late_90d / count_90d) * 1.4)
X (Unit Spread, weight 0.10)     = (distinct_units_90d - 1) / 3

Composite Score = round(100 * (0.30*V + 0.25*S + 0.20*R + 0.15*A + 0.10*X))
Risk Bands      = Low (0–34) | Watch (35–59) | High (60–79) | Critical (80–100)
```

3.4 SLA State & Logistic Breach Probability Engine

```
elapsed_ratio = (now - created_at) / (due_at - created_at)
z = -2.2 + 3.2 * elapsed_ratio + 0.8 * unit_load + 0.6 * stagnation + 0.5 * is_high_or_critical
breach_probability = 1 / (1 + exp(-z))

SLA State: "closed" (Resolved/Closed) | "breached" (now > due_at) |
           "at_risk" (elapsed_ratio >= 0.70 OR breach_probability >= 0.65) | "ok"
```

3.5 Section 5 Case Status Transition State Machine

Current Status	Permitted Next Statuses (Bureau Head & Assigned Case Officer)
New	Assigned, In Review, Escalated
Assigned	In Review, Awaiting Institution, Escalated, Resolved
In Review	Awaiting Institution, Escalated, Resolved
Awaiting Institution	In Review, Escalated, Resolved
Escalated	In Review, Awaiting Institution, Resolved
Resolved	Closed, In Review (Reopen — also permitted for Chairman)
Closed	In Review (Reopen — also permitted for Chairman)

SECTION 04

Data Schemas & REST API Specification

4.1 Core Data Entities

- **UnitRecord:** id, code ("AB" | "ACB" | "SGC" | "VC"), name, kind ("Bureau" | "Cell").
- **OfficerRecord:** id, name, unit_id, title (includes Case Officer Kavita Nair, id=6).
- **InstitutionRecord:** id, name, city, state, type, established_year, intake.
- **ComplaintRecord:** id, ref (AICTE-2026-XXXXX), institution_id, title, description, category, severity, status, unit_id, assigned_officer_id, complainant_type, created_at, updated_at, due_at, resolved_at, escalation_count, auto_confidence, cluster_id.
- **CaseLinkRecord:** id, institution_id, a_id, b_id, type ("Duplicate" | "Similar" | "Context"), similarity, resolution_status ("pending" | "confirmed" | "dismissed"), resolved_by, resolved_at.
- **AlertRecord:** id, rule ("R1" | "R2" | "R3" | "R4"), severity, institution_id, complaint_id, message, created_at, acknowledged, acknowledged_by.

4.2 REST API Endpoints (/api/*)

Method & Path	Role Access	Description
GET /api/auth/accounts	Public	Returns the 4 official role accounts and password hints.
POST /api/auth/login	Public	Validates email + password hash; returns Bearer token & UserInfo.
GET /api/me	Authenticated	Returns active SessionUser and computed RolePermissions.
GET /api/overview	Chairman, Bureau Head	Returns Chairman Dashboard or Bureau Dashboard KPIs & workload.
GET /api/complaints	All 4 Roles	Paginated list; Case Officer sorted assigned-first; Admin redacted.
POST /api/complaints/analyze	Bureau Head, Case Off.	Live TF-IDF link & risk delta preview before filing.
POST /api/complaints	Bureau Head, Case Off.	Registers complaint, routes unit, computes SLA & links.
GET /api/complaints/:id	All 4 Roles	Full dossier & timeline (Admin receives metadata-only redaction).
PATCH /api/complaints/:id	Role-Enforced	Assign officer, transition status (Sec 5 / Reopen), or add note.
POST /api/complaints/:id/escalate	Bureau Head, Case Off.	Escalates case, increments escalation_count, fires R3 alert.
GET /api/link-queue	All 4 Roles	Lists candidate Duplicate/Similar/Context links across cases.
POST /api/link-queue/:id/resolve	Bureau Head, Admin	Marks candidate link as confirmed or dismissed with audit event.
GET /api/institutions	All 4 Roles	Paginated institution directory with 5-factor risk scores.
GET /api/institutions/:id	All 4 Roles	Institution risk dossier, factor breakdown, 12w sparkline & cases.
GET /api/institutions/:id/graph	All 4 Roles	Nodes and non-dismissed link edges for SVG network topology.
GET /api/alerts	All 4 Roles	Early-warning alerts generated by rules R1, R2, R3, and R4.

POST /api/alerts/:id/acknowledge	Chairman, Bureau Head	Marks an alert as acknowledged and records actor name.